

3.4 GPT and BERT

Same block, different mask and objective: next-token versus masked tokens.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

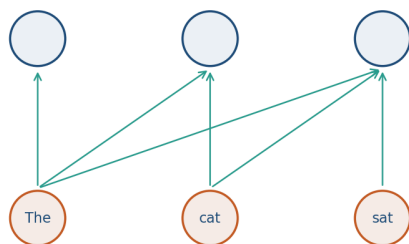
GPT and BERT are the same block with different **masks** and **objectives**. That choice is the difference between a generator and a bidirectional encoder.

1. Two attention patterns

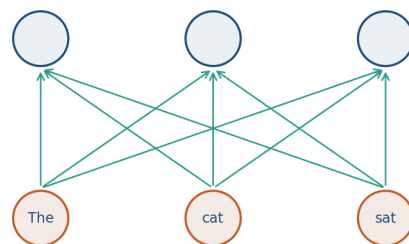
GPT (decoder-only): each token may attend to itself and the past. The pretraining job is **next-token** prediction. Sampling is ancestral: emit t , then condition on it.

BERT (encoder-only): every token may attend to every token. The pretraining job is **masked language modeling** (and originally next-sentence prediction). You cannot sample a paragraph left-to-right without extra machinery.

GPT: causal (next token)



BERT: bidirectional (masked token)



2. Objectives, in one table

	GPT	BERT
Mask	causal	none (random tokens hidden)
Loss	$p(x_t \mid x_{< t})$	$p(x_{\text{masked}} \mid x_{\text{rest}})$
Default use	generate, chat, complete	classify, tag, embed a span
Peek at the future?	no	yes, by design

Modern “BERT-like” encoders (RoBERTa, T5’s encoder, embedding models) keep the bidirectional idea. Modern LLMs are GPT-style stacks, often with extra alignment (Weeks 8–9).

3. Scaling, briefly

The same block, bigger: more layers, wider d , more heads, more data. Week 7 is scaling laws and efficiency. This week: if the mask is causal, you have a language model; if not, you have a contextual encoder. Do not fine-tune BERT as if it were GPT.

For the project: generation, RAG, and tool-use sit on a decoder. Classification into a small label set can still use an encoder, or a decoder with a prompt.

4. Practice

1. Why is BERT the wrong pretrained model if your demo must stream tokens?
2. A sentiment classifier: encoder or decoder? Give one reason for each.
3. What does a causal mask set the illegal scores to before the softmax, and why not zero?